MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1DSCPHY100 – Foundations of Physics

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **7** questions

Each question carries **2** Mark

1	What are the various problem solving strategies in physics?	[K][1]
2	Differentiate between scalar and vector quantities.	[U][1]
3	Find the displacement of a particle if its initial and final positions are 18 m and 109 m respectively.	[A][2]
4	Differentiate between acceleration and deceleration.	[K][3]
5	Why must Airshow pilots keep track of their motion relative to their audience?	[U][3]
6	Define action and reaction in the context of Newton's third law.	[K][3]
7	How does gravitational potential energy change during free fall?	[U][4]
8	Is friction classified as a conservative force? Explain.	[U][4]
9	What is the difference between positive work and negative work?	[U][4]
10	What is a comment in python programming?	[K][5]

(2x7=14)

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **6** marks

11	Which of the following is the most precise device for measuring length and explain the reasons. a) Vernier calliper with 10 divisions on the sliding scale b) Screw gauge of pitch 1mm and 100 divisions on the circular scale.	[U][1]
12	A cheetah is found at 20 m to the east of a vehicle. At time t = 0, it begins to run due east towards its prey which is at 50 m to the east of the vehicle. During the first 2.0 s of the chase, the Cheetah's x- coordinate varies with the time according to the equation $x = 20m + (5m/s^2)t^2$. Find the Cheetah's instantaneous velocity at t_1 = 1 s by taking, Δt = 0.1 s, then 0.01 s and 0.001 s.	[A][2]
13	A ball is thrown vertically upwards from the top of a tall building with velocity of 20 m/s. The height from the ground level to the point where the ball is thrown is 25.0 m. a)How high will the ball travel? b) How long does it take to reach the floor? Take g = 10 m/s2.	[An][3]
14	Describe how the principle of conservation of energy applies to a system involving gravitational potential energy	[U][4]
15	What role does the angle between force and displacement play in determining the work done?	[K][4]
16	Write a Python program to determine if a given number is positive, negative, or zero.	[A][5]

(6x4=24)

Part C

Essay Type Question

Answer any **1** question

Each question carries **12** marks

17	Describe energy diagram of stable and unstable equilibrium with examples.	[U][4]
18	Explain append, indexing and slicing with suitable examples of each.	[A][5]

(1x12=12)